

Environmental obesogens: organotins and endocrine disruption via nuclear receptor signaling.

Over the last two decades, the incidence of obesity and associated metabolic syndrome diseases has risen dramatically, becoming a global health crisis. Increased caloric intake and decreased physical activity are believed to represent the root causes of this dramatic rise. However, recent findings highlight the possible involvement of environmental obesogens, xenobiotic chemicals that can disrupt the normal developmental and homeostatic controls over adipogenesis and energy balance. Environmental estrogens, i.e. chemicals with estrogenic potential, have been reported to perturb adipogenic mechanisms using in vitro model systems, but other classes of endocrine-disrupting chemicals are now coming under scrutiny as well. Organotins represent one class of widespread persistent organic pollutants with potent endocrine-disrupting properties in both invertebrates and vertebrates. New data identify tributyltin chloride and triphenyltin chloride as nanomolar agonist ligands for retinoid X receptor (RXR alpha, RXR beta, and RXR gamma) and peroxisome proliferator-activated receptor gamma, nuclear receptors that play pivotal roles in lipid homeostasis and adipogenesis. The environmental obesogen hypothesis predicts that inappropriate receptor activation by organotins will lead directly to adipocyte differentiation and a predisposition to obesity and/or will sensitize exposed individuals to obesity and related metabolic disorders under the influence of the typical high-calorie, high-fat Western diet. The linking of organotin exposure to adipocyte differentiation and obesity opens an important new area of research into potential environmental influences on human health and disease.